BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MIRPURKHAS

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

SECTION - A

Qi. The quantity of motion In a body Is called

(A) Forts
(B) Mettle
(C) Momentum
(D) Gravitation

Qii. The direction of centrifugal force Is always

(A) Away from centre
(B) Toward MVO
(C) At centre
(D) along circumfarnace

Qiii. Nevnows Net law of motion also celled !swig

(A) Velocity
(B) Rest
(C) Inertia
(D) Force

Qiv. A 1SN force making angle of 60' with horizontal, Its vertical component wills be

(A) ISN
(B) 10N
(C) 7N
(D) 13N

Qv. A body Is said to be In neutral equilibrium If Its centre of gravity Is

(A) Al the base of body
(B) At bottom
(C) Above the body
(D) All same highet

Qvi. Which of the following is more elastic matter?

(A) Rubber
(B) Glass
(C) Steel
(D) Wood

Qvii. Which of the following Is not a unit of pressure?

(A) Pascal
(B) Bar
(C) Atmosphere
(D) Newton

Qviii. A metallic block applies force of 20N on 5cm² area. The pressure applied by block Is

(A) 100 Micro'
(B) 0.8 Mem'
(C) 0.25 Nkm'
(D) 4/N/cm1.

Qix. Gravitational force is always _____ force.

(A) Repulsive
(B) Attractive
(C) Batj (A) and(B)
(D) None of these

Qx. Critical velocity Vc = _____.

(A) g^{R}
(B) g^{IR}
(C) gR
(D) g/R

Qxi. In which process gases converts into liquid.

(A) Combustion
(B) Freezing
(C) Sublimation
(D) Bioling

Qxii. The burning candle is an example of _____ state of matter.

(A) Gas
(B) Liquid
(C) Solid
(D) All of these

Qxiii. If a machine works 20 Joule for 10 seconds then the power of machine will be:

(A) 200 watt
(B) 20 watt
(C) 2 watt
(D) 0.2 watt

Qxiv. The ratio of work done to the energy supplied Is called

(A) Force
(B) Work
(C) Power
(D) Efficiency

Qxv. The temperature at which the liquid boils is called _____

(A) Melling point
(B) Freezing point
(C) Boiling point
(D) Both (A) and (C)

Qxvi. The co efficient of volume expansion β = _____

(A) 1/2α
(B) 2α
(C) 1/3α
(D) 3α

Qxvii. Least count of Venire Calipers is

(A) 0.1mm
(B) 1mm
(C) 0.01mm
(D) 10mm

Qxviii. Physical balance is a device used to measure

(A) Weight
(B) Length
(C) Time
(D) Pressure

Qxix. 1Kilogram Is equal to _____

(A) 10^2g
(B) 10^1g
(C) 10^4g
(D) 10^9g

Qxx. The fundamental unit of length In S.I unit of measurement is

(A) Kilometer
(B) Meter
(C) Yard
(D) Foot

Qxxi. The zero error of an instrument Is error.

(A) Systematic
(B) Human
(C) Accidental
(D) All

Qxxii. Which of following is not derived physical quality

(A) Area
(B) Density
(C) Length
(D) Velocity

Qxxiii. Which of following is scalar quantity?

(A) Force
(B) Time
(C) Velocity
(D) Weight

Qxxiv. The Shortest distance between two points called:

(A) Distance
(B) Displacement
(C) Speed
(D) Velocity

SECTION - B

> Note: ALL QUESTIONS CARRY EQUAL MARKS.

Q2. Define fundamental and derived physical quantities. Give three examples each.

Q3. Prove that S = vit [illegible in source scan]

Q4. Define force of friction and describe its advantages.

Q5. If horizontal and vertical components of force are 1N and 3N respectively. Find the magnitude and direction of force.

Q6. Describe the states of equilibrium.

Q7. Define pressure and Its unit.

Q8. Distinguish between any one.

(i) Weight and Mass

(ii) Heal and Temperature

Q9. A machine performs 900 Joule of work for 30 minutes. Find the power of machine?

Q10. A Pall Is dropped from a height of 70m. Find the lime required to reach the ball at ground.

Q11. What is the contribution of Al. Haitham in the field of Physics?

Q12. Define force. How does second law of motion help in the measurement of force?

Q13. What is centripetal force? Give example of a body moving uniformly in a circle directed towards the centre.

SECTION - C

> Note: ANSWER ANY TWO (02) OF THE FOLLOWING QUESTIONS.

Q14. Define momentum and explain law of conservation of momentum.

Q15. State and explain law of universal gravitation and write its properties.

Q16. Define thermal expansion. Explain the linear expansion